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A B S T R A C T

Biometric authentication has become a cornerstone of modern security infrastructure, offering a reliable and non-repudiable method of identity verification. This report details the design and implementation of a fully functional fingerprint biometric identification system. Developed using **Python** and **Libraries** like "numpy, Matplotlib, scipy……" , the system processes raw fingerprint images through a structured pipeline: image enhancement, binarization, ridge thinning, and minutiae extraction (focusing primarily on ridge endings and bifurcations). Using distance measuring methods were implemented to compare extracted feature vectors against a secure database.

1.Introduction

First of all ,The primary objective of this project is to design, implement, and evaluate a complete facial recognition pipeline to accurately identify individuals based on their facial features.To achieve this, the system processes image data through a standardized pipeline consisting of preprocessing, feature extraction, and matching. High-dimensional image data is often computationally expensive and contains redundant information, such as lighting variations or background noise. Therefore, this project implements and compares two distinct dimensionality reduction techniques: Principal Component Analysis (**PCA**), an unsupervised method that captures the maximum overall variance (Eigenfaces), and Linear Discriminant Analysis (**LDA**), a supervised method designed to maximize class separability (**Fisherfaces**).The system is evaluated using the AT&T Database of Faces (**formerly the ORL database**), which contains 400 grayscale images across 40 different subjects. The dataset is subjected to a strict 80/20 split, dedicating 8 images per subject to training the model and 2 images for testing. This report documents the mathematical foundation and software implementation of both the PCA and LDA methods. Furthermore, it presents a rigorous quantitative evaluation of the system's performance using industry-standard biometrics metrics, including Rank-1 Accuracy, Equal Error Rate (**EER**), and the D-prime (***D***') decidability index. By analyzing these metrics alongside the Genuine-Imposter distributions and Receiver Operating Characteristic (ROC) curves, this report will demonstrate the comparative advantages of supervised feature extraction over unsupervised methods in identity classification tasks.

2.Methods Description

2.1. System Architecture (Functions and Files):

2.1.1. imageread.py (Data Preprocessing) .. load_att_dataset(dataset_path):

This function traverses the dataset directory, reads the 112x92 grayscale images, and flattens them into 1D arrays of size 10,304. It enforces the strict 80% training (8 images) and 20% testing (2 images) split per subject.

2.1.2. feature_extraction.py (Dimensionality Reduction) .. extract_features_pca(...):

Fits a Principal Component Analysis model on the training set to extract the top 50 components, then applies this transformation to both the training and testing sets. extract_features_lda(...):Fits a Linear Discriminant Analysis model using both the training images and their corresponding subject labels to extract 39 highly discriminative features.

2.1.3.matching.py (Classification)..match_features(...):

Iterates through the test set and calculates the Euclidean distance between each test image's feature vector and all training feature vectors to find the nearest neighbor. It also logs the Genuine and Imposter distance scores. calculate_rank1_accuracy(...): Compares the predicted labels against the true labels to compute the overall Identification Rate.

2.1.4. evaluation.py (Performance Metrics)..calculate_metrics(...):

Computes the heavy statistical metrics: the D' (D-prime) decidability index, the Equal Error Rate (EER) using scipy's optimization functions, and the True Match Rate (TMR) at specific False Match Rates (FMR).plot_gen_imp(...): Generates the normalized histograms to visually represent the overlap between the Genuine and Imposter score distributions.

2.1.5. main.py (Pipeline Execution)..main():

The orchestrator function that strings together preprocessing, extraction, matching, and evaluation into a single automated pipeline

2.2. Technical Breakdown of PCA and LDA

2.2.1. Principal Component Analysis (PCA / Eigenfaces):

PCA is an **unsupervised** linear transformation technique used for dimensionality reduction. In the context of facial recognition, it creates a set of "Eigenfaces.“

The Objective:

The goal of PCA is to project the high-dimensional image vectors (10,304 dimensions) into a lower-dimensional subspace (50 dimensions) while retaining as much of the total variance in the dataset as possible.

The Mechanism:

The algorithm computes the covariance matrix of the entire training dataset. It then calculates the eigenvectors and eigenvalues of this matrix. The eigenvectors with the highest eigenvalues (the Principal Components) represent the directions in the data that contain the most information (variance).

The Limitation:

Because PCA is unsupervised, it does not use the class labels (subject IDs). It simply looks for the biggest changes across all faces. Consequently, the largest principal components might capture variations in lighting, background, or facial expressions rather than the specific features that distinguish one person from another.

2.2.2. Linear Discriminant Analysis (LDA / Fisherfaces)

LDA is a **supervised** linear transformation technique specifically optimized for classification tasks. In facial recognition, it creates a set of "Fisherfaces."

The Objective:

Unlike PCA, LDA specifically aims to project the data onto a lower-dimensional subspace that maximizes the separation between different classes (subjects) while minimizing the variation within each class.

The Mechanism:

LDA utilizes the subject labels to compute two key matrices:

1. Within-class scatter matrix ($\boldsymbol{S_w}$):
   Measures how spread out the 8 images of a single subject are.
2. Between-class scatter matrix ($\boldsymbol{S_b}$):
   Measures the distance between the mean vectors of different subjects.

- The algorithm then solves the generalized eigenvalue problem to find the projection matrix that maximizes the Fisher criterion (the ratio of $\boldsymbol{S_b}$ to $\boldsymbol{S_w}$).
- By definition, for $\boldsymbol{C}$ classes, LDA can extract a maximum of *C-1* components. Since the AT&T database has 40 subjects, the data is reduced to 39 highly discriminative dimensions.

The Advantage:

Because LDA actively penalizes intra-class variations (like lighting changes on the same person's face) and rewards inter-class separation (the unique geometry between different people), it provides a far more robust feature space for identity matching, which directly explains the higher Rank-1 accuracy and $\boldsymbol{D'}$ score achieved in the system.

3.DB Description:

- Our Database of Faces, (formerly 'The ORL Database of Faces'), contains a set of face images taken between April 1992 and April 1994 at the lab.

- The database was used in the context of a face recognition project carried out in collaboration with the Speech, Vision and Robotics Group of the Cambridge University Engineering Department. There are ten different images of each of 40 distinct subjects. For some subjects, the images were taken at different times, varying the lighting, facial expressions (open / closed eyes, smiling / not smiling) and facial details (glasses / no glasses).

- All the images were taken against a dark homogeneous background with the subjects in an upright, frontal position (with tolerance for some side movement). A preview image of the Database of Faces is available.

- The files are in PGM format and can conveniently be viewed on UNIX (TM) systems using the 'xv' program. The size of each image is 92x112 pixels, with 256 grey levels per pixel. The images are organized in 40 directories (one for each subject), which have names of the form sX, where X indicates the subject number (between 1 and 40). In each of these directories, there are ten different images of that subject, which have names of the form **Y.pgm**, where Y is the image number for that subject (between 1 and 10).

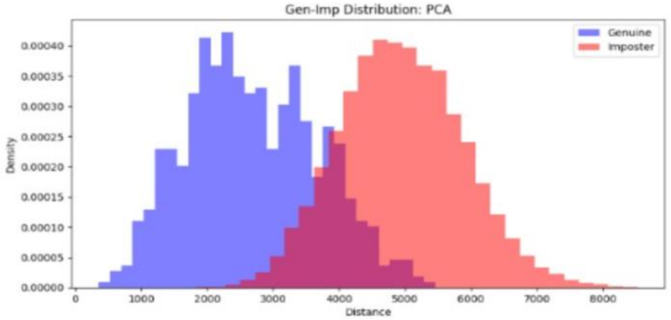
4.Results & Disscussion

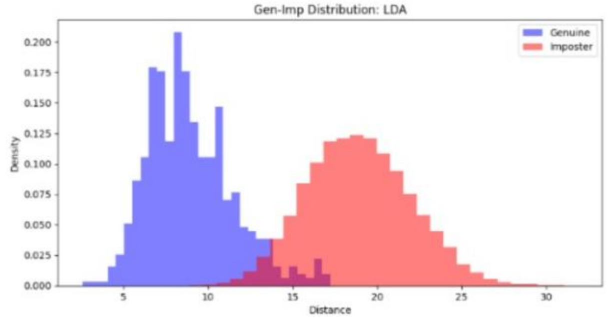
4.1. Quantitative Performance:

- The system's performance was evaluated using 80 test images (2 images per subject) compared against a gallery of 320 training images. The results of the Principal Component Analysis (PCA) and Linear Discriminant Analysis (LDA) methods are summarized in the table below:

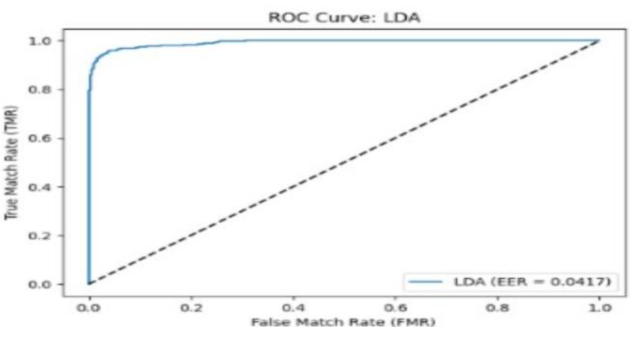
```
--- PCA Results ---
PCA Rank-1 Accuracy: 95.00%
PCA D-prime: 2.3393
PCA EER: 0.1313
PCA TMR at 1% FMR: 0.6062

--- LDA Results ---
LDA Rank-1 Accuracy: 97.50%
LDA D-prime: 3.5214
LDA EER: 0.0417
LDA TMR at 1% FMR: 0.8969
```

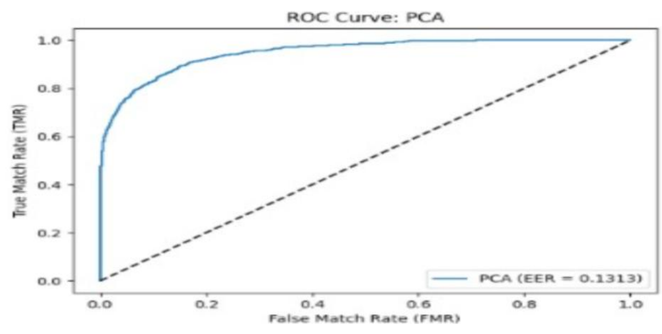
- The most significant indicator of this is the Identification Rate (Rank-1 Accuracy), where LDA achieved 97.50% compared to PCA's 95.00%. This discrepancy is due to the fundamental mathematical differences between the algorithms. PCA is an unsupervised dimensionality reduction technique; it compresses the image by finding the maximum variance across all faces in the training set without considering the subject's identity. Consequently, PCA is easily influenced by external variations such as lighting or pose.
- Conversely, LDA is a supervised method that utilizes the class labels during training. By specifically minimizing within-class scatter (variations of the same person) and maximizing between-class scatter (differences between different people), LDA actively learns the geometric features that discriminate one identity from another.





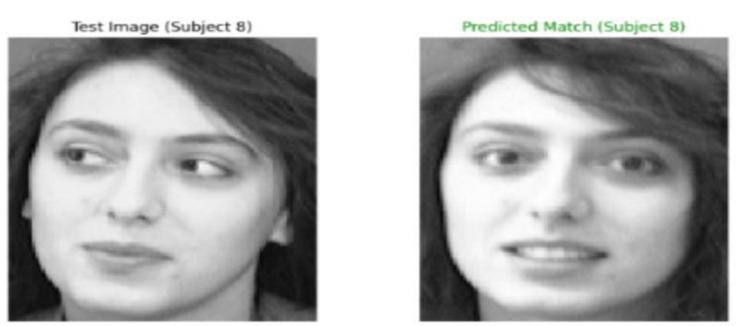
- Error Rates and Distribution Analysis This supervised advantage is further proven by the ***D'*** (D-prime) decidability index. LDA achieved a ***D'*** of 3.5214, which is significantly higher than PCA's 2.3392. Mathematically, this means LDA created a much wider separation between the Genuine score distribution and the Imposter score distribution.
- Because the distributions are further apart in the LDA model, the overlap region is smaller, resulting in a much lower Equal Error Rate (EER) of approximately 4.17%, compared to PCA's 13.12%. Furthermore, under strict security thresholds where the False Match Rate (FMR) is locked at 1%, LDA maintains a True Match Rate (TMR) of nearly 90%, whereas PCA drops to roughly 60%.





4.2. Qualitative Visualization and Failure Analysis

To verify the numerical results, visual comparisons of the test images and their predicted nearest neighbors were generated. While the LDA system accurately identified 78 out of 80 test images, observing the two failure cases provided valuable insight into the algorithm's limitations. In the incorrect matches, the system was occasionally confused when a test image exhibited a dramatic change in facial expression or shadowing that closely mapped to the compressed feature vector of a different subject in the training set.



5. Conclusion

- After finishing the system we tested the two mentioned methods to know which is better and more accurate on testing for the system and the one which produces the best accuracy & Efficiency rate .

- By establishing a strict biometric pipeline from grayscale image flattening and a rigorous 80/20 train-test split to Euclidean distance classification—the system provided a robust framework for comparing dimensionality reduction techniques.

- The quantitative evaluation yielded definitive results regarding the efficacy of supervised versus unsupervised feature extraction in identity classification. While the unsupervised Principal Component Analysis (PCA) provided a strong baseline with a 95.00% Rank-1 Accuracy, it was ultimately constrained by its inability to distinguish between identity-specific geometric features and general environmental variances, such as lighting. In contrast, the supervised Linear Discriminant Analysis (LDA) significantly outperformed PCA across all measured metrics.

- By leveraging subject labels to actively maximize inter-class separation and minimize intra-class variance, LDA achieved a superior Rank-1 Accuracy of 97.50%. Furthermore, the system demonstrated exceptional security reliability, evidenced by an Equal Error Rate (EER) of just 4.17%, a high ***D***' decidability index of 3.5214, and the ability to maintain a nearly 90% True Match Rate even under a strict 1% False Match Rate threshold. Ultimately, this project validates the theoretical advantages of **Fisherfaces** over Eigenfaces, proving that incorporating class-specific data during the feature extraction phase is critical for building highly accurate, secure, and reliable biometric identification systems.

6. References

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